Parameters

Pulley Material - Aluminum

Belt Material - Polyurethane Rubber

μ - 0.35

SF -3

Center distance - 4in

Pulley diameter - 3in

Pulley radius - 1.5in

Operating speed -1,000 rpm

Intral State Tension - 41bf

Belt yield strength - 471 psi

Knowns

Center distance (C) 4in

Pulley Ø (D) 3in

Pulley radius 1.5in

Rotational Speed (N) 1000 rpm

Static belt Tension (T_0) 41~~bf~~ lbf

Belt yield strength (S_y) 471 psi

SF 03

Unknowns

T_1 T_2

F_t Locter Ø

F_t Power

μ d min Ø

$$\frac{T_1}{T_2} = e^{\mu\theta} \qquad \theta = \pi \text{ rad} \qquad \frac{T_1}{T_2} = e^{0.3\pi} \qquad \frac{T_1}{T_2} = 3.004$$

$$T_1 + T_2 = 2T_0 \qquad T_1 + T_2 = 8 \qquad T_1 = 3.004\,T_2$$

$$3.004\,T_2 + T_2 = 8$$

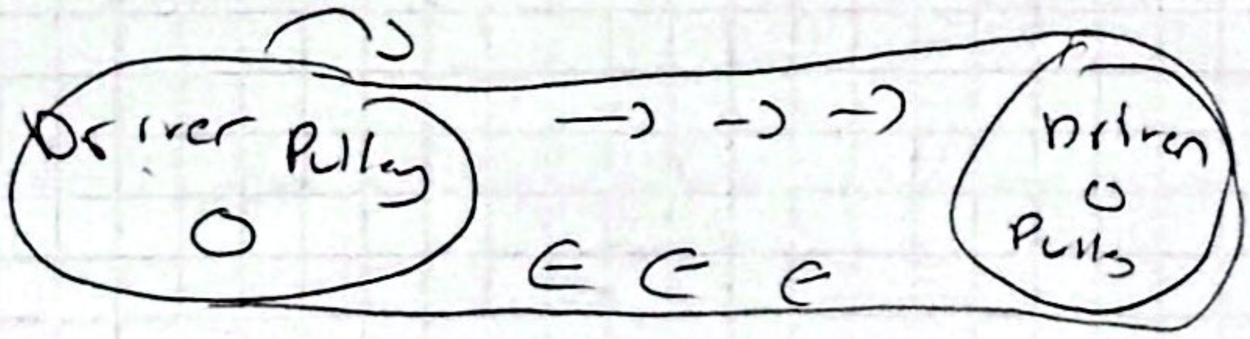
$$T_2 = 1.9976\text{lbf}$$

$$T_1 = 6.003$$

$$T_1 - T_2 = 4.0064\text{ lbf}$$

$$T = (T_1 - T_2)r$$

$$T = (6.003 - 1.997)(1.5) = 6.009\text{ lbf}\cdot\text{in}$$



$T_1 - T_2$, R_1 R_2

$$n = \frac{S_y}{\sigma_{allow}}$$

$$\sigma_{allow} = \frac{S_y}{n} = \frac{471}{3} = 157\text{ psi}$$

$$\sigma = \frac{F}{A} \qquad A = \frac{F}{\sigma} \qquad A_{min} = \frac{T_1}{\sigma_{allow}} = \frac{6.003}{157} = 0.03824\text{ in}^2$$

$$A = \frac{\pi d^2}{4}$$

$$d = \sqrt{\frac{4A}{\pi}} = \sqrt{\frac{4(0.03824)}{\pi}} = 0.2206 \text{ in dmin}$$

Selected belt = 17.5 , 3044K606 Belt

Actual values

$$A = \frac{\pi (0.25)^2}{4} = 0.04909 \text{ in}$$

$$\sigma = \frac{T_1}{A} = \frac{6.003}{0.04909} = 122.24 \text{ psi}$$

$$n = \frac{S_y}{\sigma} = \frac{471}{122.24} = 3.85$$

$$L = 2C + \pi D = 2(4) + \pi(3) = 17.425 \text{ in}$$

Using 1060

Hardened Steel

Knowns

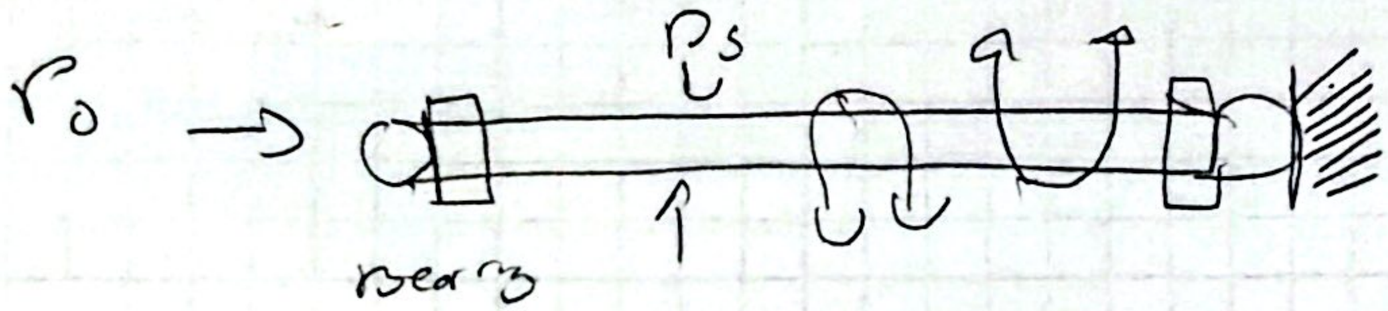
Transmitted Torque	T		6.009 lbf in
Pulley Rotational Speed		N	1,000 rpm
Pulley Radius		r	1.5 in
Tight side Tension		T_1	6.003 lbf
Loose side Tension		T_2	1.997 lbf

Shaft material — AISI 1060 Hardened steel

Safety Factor n — 3

Unknown:

S_y τ_{allow} A_{min} d_{min} τ_{max}



$$\bar{F}_R = \bar{F}_1 + \bar{F}_2$$

$$= 6.003 + 1.997 = 8.000 \text{ lbs}$$

$$S_y = 97{,}000 \text{ psi}$$

$$\tau_{max} = \frac{Tc}{J} = \frac{T(d/2)}{\pi d^4/32} = \frac{16T}{\pi d^3}$$

$$\tau_{yield} = \frac{S_y}{2} = \frac{S_y}{2n} = \frac{97{,}000}{2(3)}$$

$$= 16166.67 \text{ psi}$$

$$\tau_{max} \leq \tau_{allow}$$

$$\frac{16T}{\pi d^3} \leq \frac{S_y}{2n}$$

$$d^3 \geq \frac{32n}{\pi S_y}$$

$$d_{min} = \sqrt[3]{\frac{32 n_7}{\pi S_y}}$$

$$= \sqrt[3]{\frac{32(3)(6.009)}{\pi (97000)}} = 0.1237 \text{ in}$$

$$A_{min} = \frac{\pi d^4}{4} = \frac{\pi (0.1237)^2}{4} =$$

$$0.01202 \text{ in}^2$$

$d_{min} = 0.1237 \text{ in}$ round up to select $= 0.125$ or $\frac{1}{8}$

min $\varnothing = 0.125$ due to pure torsion.

Journal Bearing

- Porous Bronze material

Knowns

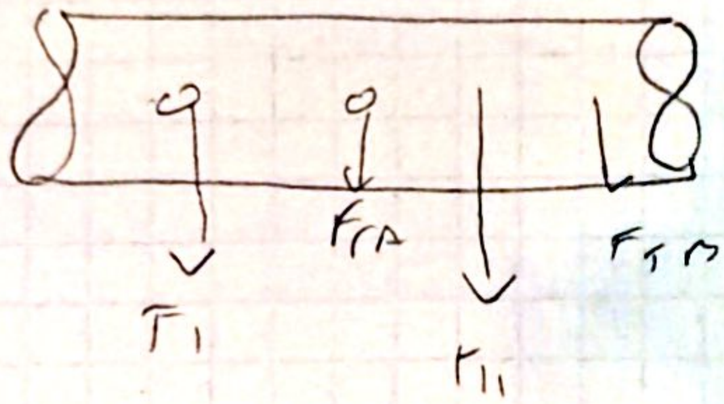
Shaft $\varnothing$	0.125
RPM	1,000
Tight side Tension	6.003 lbf
Loose side Tension	1.997 lbf
Load	8 Lbf
Bearing type	Journal Bearing

$$\sum F_R = T_1 + T_2 = 8 \text{ lbf}$$

$$R_A = R_B = \frac{F_R}{2} = R_A = 4 \text{ Lbf}$$

Unknown

P V PV L d Material



$\Sigma F_y = 0$

$R_A + R_B - F_R = 0$

$R_A = 4$

Preliminary dimension

Inside ⌀ 0.125 in

L 0.250 in

Load 41bf

$\frac{L}{d} = \frac{0.250}{0.0025} = 2$

$P = \frac{F}{A}$ $A = Ld$ $P = \frac{F}{Ld}$

$V = \frac{\pi d N}{12}$ $PV = P(V)$

$P = \frac{F}{Ld} = \frac{4}{(.250)(.125)} = 128 \text{ psi}$

$V = \frac{\pi d N}{12} = \frac{\pi (0.125)(1000)}{12} = 32.724 \text{ ft/min}$

$PV = (128)(32.72) = 4188.16 \frac{\text{psi} \cdot \text{ft}}{\text{min}}$

Porous bronze

	Calculated	Porous Bronze Limit	
Pressure	128 PSI	4,500 psi	✓
V	32.72 ft/min	1,500 ft/min	✓
P V	4,188.16	50,000	✓

$T_{max} = 160°F$

Frame

